HW 3

Kevin Lin

10/15/2026

1

Let $S = \sum_{i=1}^n X_i$, and $S_\epsilon = \sum_{i=1}^n \epsilon X_i$. The Rademacher symmetrization lemma states that if $X_1, \ldots, X_n$ are independent mean zero random variables and $F : [0, \infty) \to \mathbb{R}$ is convex and increasing, then for $\epsilon_1, \ldots, \epsilon_n$ independent Rademacher random variables, we have:

$$E\left[F\left(\frac{1}{2}\left|\sum_{i=1}^n \epsilon_i X_i\right|\right)\right] \leq E\left[F\left(\left|\sum_{i=1}^n X_i\right|\right)\right] \leq E\left[F\left(2\left|\sum_{i=1}^n \epsilon_i X_i\right|\right)\right]$$

We will use the definition of a subgaussian norm:

$$||Y||_{\psi_2} = \inf\{K > 0 : E[\exp(Y^2/K^2)] \leq 2\}$$

First, define an upper bound. Let $K_\epsilon = ||S_\epsilon||_{\psi_2}$. Then, for any $K > K_\epsilon$, we have $E[\exp(S_\epsilon^2/K^2)] \leq 2$. Choose $F(t) = \exp(t^2/(2K)^2)$, which is convex and increasing. Then, by the Rademacher symmetrization lemma (right hand side), we have:

$$E\left[\exp\left(\frac{S^2}{(2K)^2}\right)\right] \leq E\left[\exp\left(\frac{(2|S_\epsilon|)^2}{(2K)^2}\right)\right] = E\left[\exp\left(\frac{S_\epsilon^2}{K^2}\right)\right] \leq 2$$

Then, by the definition of the subgaussian norm, we have $||S||_{\psi_2} \leq 2K$. Since this holds for every $K > K_\epsilon$, we have $||S||_{\psi_2} \leq 2||S_\epsilon||_{\psi_2}$, and we take $C = 2$.
Now, define a lower bound. Let $K = ||S||_{\psi_2}$. Then, for any $L > K$, we have $E[\exp(S^2/L^2)] \leq 2$. Choose $F(t) = \exp(t^2/L^2)$, which is convex and increasing. Then, by the Rademacher symmetrization lemma (left hand side), we have:

$$E\left[\exp\left(\frac{(1/2)|S_\epsilon|^2}{L^2}\right)\right] = E\left[\exp\left(\frac{|S_\epsilon|^2}{(2L)^2}\right)\right] \leq E\left[\exp\left(\frac{S^2}{L^2}\right)\right] \leq 2$$

Then, $||S_\epsilon||_{\psi_2} \leq 2L$. Since this holds for every $L > K$, we have $||S_\epsilon||_{\psi_2} \leq 2||S||_{\psi_2}$, which rearranges to $\frac{1}{2}||S_\epsilon||_{\psi_2} \leq ||S||_{\psi_2}$, and we take $c = 1/2$. Thus,

we can combine the two bounds to get:

$$\frac{1}{2}||S_\epsilon||_{\psi_2} \leq ||S||_{\psi_2} \leq 2||S_\epsilon||_{\psi_2}$$

$$\frac{1}{2}\left\|\sum_{i=1}^{n}\epsilon_i X_i\right\|_{\psi_2} \leq \left\|\sum_{i=1}^{n} X_i\right\|_{\psi_2} \leq 2\left\|\sum_{i=1}^{n}\epsilon_i X_i\right\|_{\psi_2}$$

where $c = 1/2$ and $C = 2$.

2

(a) We substitute β^* into the Dantzig selector and get $\|\frac{1}{n}X^T(y - X\beta^*)\|_\infty$. Since the linear model is $y = X\beta^* + \epsilon$, we can substitute this in as well to get $\|\frac{1}{n}X^T(X\beta^* + \epsilon - X\beta^*)\|_\infty = \|\frac{1}{n}X^T\epsilon\|_\infty \leq \lambda$. The last inequality holds by the given assumption, thus β^* is a feasible solution. By definition, $\hat{\beta}_D = \arg\min_{\beta\in\mathcal{F}} \|\beta\|_1$, where the feasible set $\mathcal{F} = \{\beta \in \mathbb{R}^p : \|\frac{1}{n}X^T(y - X\beta)\|_\infty \leq \lambda\}$. We know that $\beta^* \in \mathcal{F}$, and since $\hat{\beta}_D$ is the minimizer of the ℓ_1 norm over the feasible set, its objective value has to be less than or equal to the objective value at any other feasible vector, including β^*, thus, we have $\|\hat{\beta}_D\|_1 \leq \|\beta^*\|_1$.

(b) Let $\Delta = \hat{\beta}_D - \beta^*$, $S = \text{supp}(\beta^*)$, and $|S| = \kappa$. We can rewrite $\hat{\beta}_D = \beta^* + \Delta$, and from (a), we get $\|\beta^* + \Delta\|_1 \leq \|\beta^*\|_1$. Because $S = \text{supp}(\beta^*)$, we know that $\beta^*_{S^c} = 0$, and thus we can rewrite the ℓ_1 norm as $\|\beta^* + \Delta\|_1 = \|\beta^*_S + \Delta_S\|_1 + \|\beta^*_{S^c} + \Delta_{S^c}\|_1 = \|\beta^*_S + \Delta_S\|_1 + \|\Delta_{S^c}\|_1$. Thus, $\|\beta^*_S + \Delta_S\|_1 + \|\Delta_{S^c}\|_1 \leq \|\beta^*\|_1$. With the reverse triangle inequality, we have $\|\beta^*_S + \Delta_S\|_1 \geq \|\beta^*_S\|_1 - \|\Delta_S\|_1$. Substituting this into the previous inequality, we have $\|\beta^*_S\|_1 - \|\Delta_S\|_1 + \|\Delta_{S^c}\|_1 \leq \|\beta^*\|_1$. Then, we get $-\|\Delta_S\|_1 + \|\Delta_{S^c}\|_1 \leq 0$, which rearranges to $\|\Delta_{S^c}\|_1 \leq \|\Delta_S\|_1$, proving the cone condition.

We can bound the ℓ_1 norm since S and S^c partition the coordinates, where $\|\Delta\|_1 = \|\Delta_S\|_1 + \|\Delta_{S^c}\|_1 \leq \|\Delta_S\|_1 + \|\Delta_S\|_1 = 2\|\Delta_S\|_1$. Then, we can bound $\|\Delta_S\|_1$ using the Cauchy-Schwarz inequality and the fact that $|S| = \kappa$:

$$\begin{aligned}\|\Delta_S\|_1 &= \sum_{i\in S}|\Delta_i| \\ &\leq \sqrt{\sum_{i\in S} 1^2}\sqrt{\sum_{i\in S}|\Delta_i|^2} \\ &= \sqrt{\kappa}\|\Delta_S\|_2\end{aligned}$$

$\|\Delta_S\|_2 \leq \|\Delta\|_2$, because the squared Euclidean norm over the subset of coordinates cannot exceed the squared Euclidean norm over all coordi-

nates. Thus, combining the inequalities, we have:

$$\begin{aligned}\|\Delta\|_1 &\leq 2\|\Delta_S\|_1 \\ &\leq 2\sqrt{\kappa}\|\Delta_S\|_2 \\ &\leq 2\sqrt{\kappa}\|\Delta\|_2\end{aligned}$$

Therefore, $\|\hat{\beta}_D - \beta^*\|_1 \leq 2\sqrt{\kappa}\|\hat{\beta}_D - \beta^*\|_2$.

(c) Starting with the Dantzig selector $\|\frac{1}{n}X^T(y - X\hat{\beta}_D)\|_\infty \leq \lambda$, we can substitute y:

$$\begin{aligned}\|\frac{1}{n}X^T(y - X\hat{\beta}_D)\|_\infty &= \|\frac{1}{n}X^T(X\beta^* + \epsilon - X\hat{\beta}_D)\|_\infty \\ &= \|\frac{1}{n}X^T(\epsilon - X(\hat{\beta}_D - \beta^*))\|_\infty \\ &= \|\frac{1}{n}X^T\epsilon - \frac{1}{n}X^TX\Delta\|_\infty \\ &\leq \lambda\end{aligned}$$

Rearrange the inequality and then take the ℓ_∞ norm with the triangle inequality:

$$\begin{aligned}\frac{1}{n}X^TX\Delta &= \frac{1}{n}X^T\epsilon - \frac{1}{n}X^T(y - X\hat{\beta}_D) \\ \|\frac{1}{n}X^TX\Delta\|_\infty &= \|\frac{1}{n}X^T\epsilon + \frac{1}{n}X^T(y - X\hat{\beta}_D)\|_\infty \\ &\leq \|\frac{1}{n}X^T\epsilon\|_\infty + \|\frac{1}{n}X^T(y - X\hat{\beta}_D)\|_\infty \\ &\leq \lambda + \lambda = 2\lambda\end{aligned}$$

The last inequality follows from the given noise assumption and the Dantzig selector constraint. Thus, we have $\|\frac{1}{n}X^TX(\hat{\beta}_D - \beta^*)\|_\infty \leq 2\lambda$.

Using the definition of the squared Euclidean norm, we have:

$$\begin{aligned}\frac{1}{n}\|X\Delta\|_2^2 &= \frac{1}{n}(X\Delta)^\top(X\Delta) \\ &= \frac{1}{n}\Delta^\top X^\top X\Delta \\ &= \Delta^\top\left(\frac{1}{n}X^\top X\Delta\right)\end{aligned}$$

We can use Holder's inequality for the ℓ_1 and ℓ_∞ norms:

$$\begin{aligned}
|a^\top b| &\leq \|a\|_1 \|b\|_\infty \\
a = \Delta, b &= \frac{1}{n} X^\top X \Delta \\
\frac{1}{n}\|X\Delta\|_2^2 &= \Delta^\top \left(\frac{1}{n} X^\top X \Delta\right) \\
&\leq |\Delta^\top \left(\frac{1}{n} X^\top X \Delta\right)| \\
&\leq \|\Delta\|_1 \left\| \frac{1}{n} X^\top X \Delta \right\|_\infty \\
&\leq \|\Delta\|_1 (2\lambda) \\
&= 2\lambda \|\Delta\|_1
\end{aligned}$$

Thus, $\frac{1}{n}\|X(\hat{\beta}_D - \beta^*)\|_2^2 \leq 2\lambda \|\hat{\beta}_D - \beta^*\|_1$.

(d) From (b) we know that $\|\Delta_{S^c}\|_1 \leq 3\|\Delta_S\|_1$. Since $\|\Delta_S\|_1 \leq 3\|\Delta_S\|_1$, we also have $\|\Delta_{S^c}\|_1 \leq 3\|\Delta_S\|_1$, and thus Δ satisfies the cone condition required by the restricted eigenvalue condition. Then, we can apply the restricted eigenvalue condition to get $\frac{1}{n}\|X\Delta\|_2^2 \geq c\|\Delta\|_2^2$. From (c), we know that $\frac{1}{n}\|X\Delta\|_2^2 \leq 2\lambda\|\Delta\|_1$. Combining these two inequalities, we have:

$$\begin{aligned}
c\|\Delta\|_2^2 \leq \frac{1}{n}\|X\Delta\|_2^2 &\leq 2\lambda\|\Delta\|_1 \\
c\|\Delta\|_2^2 &\leq 2\lambda\|\Delta\|_1
\end{aligned}$$

Again, from (b), we have $\|\Delta\|_1 \leq 2\sqrt{\kappa}\|\Delta\|_2$. Substituting this into the previous inequality, we have:

$$\begin{aligned}
c\|\Delta\|_2^2 &\leq 2\lambda\|\Delta\|_1 \\
&\leq 2\lambda(2\sqrt{\kappa}\|\Delta\|_2) \\
&= 4\lambda\sqrt{\kappa}\|\Delta\|_2 \\
\|\Delta\|_2 &\leq \frac{4\lambda\sqrt{\kappa}}{c} \\
\|\Delta\|_1 &\leq 2\sqrt{\kappa}\|\Delta\|_2 \\
&\leq 2\sqrt{\kappa}\left(\frac{4\lambda\sqrt{\kappa}}{c}\right) \\
&= \frac{8\lambda\kappa}{c}
\end{aligned}$$

Therefore, $\|\hat{\beta}_D - \beta^*\|_1 \leq \frac{8}{c}\kappa\lambda$ and if we let $C = 8/c > 0$, we can conclude that $\|\hat{\beta}_D - \beta^*\|_1 \leq C\kappa\lambda$, and thus, the Dantzig selector achieves the same ℓ_1 error rate as KASSO.

HW3_P3

October 22, 2026

1 3

1.1 (a)

[1]:
```
import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(seed=0)
p, sigma = 1000, 1.0
n_values = np.array([50, 100, 200, 400, 800, 1600])
trials = 100
results = []

for n in n_values:
    measurements = []
    for _ in range(trials):
        X = rng.normal(0, 1, size=(n, p))
        epsilon = rng.normal(0, sigma, size=n)
        measurements.append(np.max(np.abs(X.T @ epsilon / n)))
    measurements = np.array(measurements)
    lam0 = sigma * np.sqrt(2 * np.log(p) / n)
    results.append((n, measurements.mean(), measurements.std(ddof=1),
                    measurements.std(ddof=1) / np.sqrt(trials), lam0,
                    measurements.mean() / lam0))

results = np.array(results)
slope = np.polyfit(np.log(results[:, 0]), np.log(results[:, 1]), 1)[0]
results, slope
```

[1]:
```
(array([[5.00000000e+01, 4.88022862e-01, 6.42583230e-02, 6.42583230e-03,
         5.25652177e-01, 9.28414042e-01],
        [1.00000000e+02, 3.43539483e-01, 3.68017559e-02, 3.68017559e-03,
         3.71692219e-01, 9.24257935e-01],
        [2.00000000e+02, 2.43493566e-01, 2.70290778e-02, 2.70290778e-03,
         2.62826088e-01, 9.26443671e-01],
        [4.00000000e+02, 1.73718932e-01, 1.70579842e-02, 1.70579842e-03,
         1.85846109e-01, 9.34746132e-01],
        [8.00000000e+02, 1.21766597e-01, 1.32863779e-02, 1.32863779e-03,
```

```
        1.31413044e-01, 9.26594443e-01],
       [1.60000000e+03, 8.64825114e-02, 8.51083136e-03, 8.51083136e-04,
        9.29230547e-02, 9.30689500e-01]]),
 np.float64(-0.4988155200940045))
```

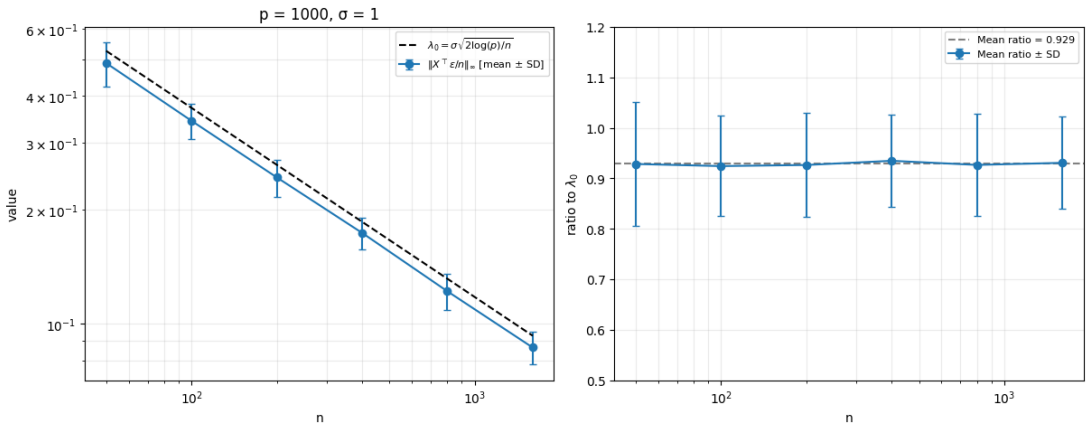
[5]:
```
from matplotlib.ticker import ScalarFormatter

fig, axes = plt.subplots(1, 2, figsize=(12, 4.7), layout="constrained")
n, avg, sd, se, lam, ratio = results.T

axes[0].errorbar(n, avg, yerr=sd, fmt="o-", capsize=3,␣
 ↪label=r"$\|X^\top\varepsilon/n\|_\infty$ [mean $\pm$ SD]")
axes[0].plot(n, lam, "k--", label=r"$\lambda_0=\sigma\sqrt{2\log(p)/n}$")
axes[0].set(xscale="log", yscale="log", xlabel="n",
            ylabel="value", title="p = 1000,   = 1")
axes[0].grid(True, which="both", alpha=.25)
axes[0].legend(fontsize=8)

axes[1].errorbar(n, ratio, yerr=sd / lam, fmt="o-", capsize=3, label="Mean␣
 ↪ratio ± SD")
axes[1].axhline(np.mean(ratio), linestyle="--", color="gray", label=f"Mean␣
 ↪ratio = {np.mean(ratio):.3f}")
axes[1].set(xscale="log", xlabel="n", ylabel=r"ratio to $\lambda_0$")
axes[1].set_ylim(.5, 1.2)
axes[1].grid(True, which="both", alpha=.25)
axes[1].legend(fontsize=8)

plt.show()
```



The fitted slope of the log-log plot is roughly -0.5. The ratio hovers around a constant 0.92. From

lecture 7, the union bound over Gaussian tails shows that:

$$\left\|\frac{1}{n}\epsilon^\top X\right\|_\infty \leq C\sigma\sqrt{\frac{\log p}{n}}$$

for some constant $C > 0$, and then chooses $\lambda = 2C\sigma\sqrt{\frac{\log p}{n}}$. This shows that noise level scales as

$$\left\|\frac{1}{n}\epsilon^\top X\right\|_\infty = O_p\left(\sigma\sqrt{\frac{\log p}{n}}\right)$$

. We have $\lambda_0 = \sigma\sqrt{\frac{2\log p}{n}}$. Both expressions have the same dependence on n and p. Since $p = 1000$, $\sigma = 1$ are fixed, both scale to $n^{-1/2}$. This explains why the experimentally observed ratio remains approximately constant at around 0.92 while n increases. The constant $\sqrt{2}$ in λ_0 is asymptotically sharp for the maximum of independent Gaussian variables, but is slightly conservative for finite p. Our observed ratio indicates that λ_0 slightly overestimates the expected noise level when $p = 1000$.

1.2 (b)

[2]:
```
from scipy.optimize import linprog
from joblib import Parallel, delayed
from threadpoolctl import threadpool_limits

rng = np.random.default_rng(seed=0)
T = 20
p, sigma = 256, 0
n_values = np.arange(8, 257, 8)
k_values = np.arange(1, 64, 2)
```

[ ]:
```
# default data generation

success = np.zeros((len(n_values), len(k_values)), dtype=float)
for i, n in enumerate(n_values):
    for j, k in enumerate(k_values):
        for trial in range(T):
            X = rng.standard_normal((n, p))
            beta = np.zeros(p)
            support = rng.choice(p, size=k, replace=False)
            beta[support] = rng.choice([-1.0, 1.0], size=k)
            y = X @ beta
            # beta = beta_plus - beta_minus, beta_plus, beta_minus >= 0
            # Objective min 1^T beta_plus + 1^T beta_minus, subject to X beta =␣
 ↪y
            fit = linprog(np.ones(2*p), A_eq=np.concatenate([X, -X], axis=1),
                          b_eq=y, bounds=(0, None), method='highs')
            if fit.success:
                beta_hat = fit.x[:p] - fit.x[p:]
                if np.linalg.norm(beta_hat-beta) <= 1e-4*np.linalg.norm(beta):
```

```
                    success[i,j] += 1
```

[ ]:
```
# parallelized

# Run all T trials for one (n, k) configuration
def run_config(i, j, n, k, seed):
    rng = np.random.default_rng(seed)
    successes = 0

    with threadpool_limits(limits=1):
        for trial in range(T):
            X = rng.standard_normal((n, p))
            beta = np.zeros(p)
            support = rng.choice(p, size=k, replace=False)
            beta[support] = rng.choice([-1.0, 1.0], size=k)
            y = X @ beta

            # beta = beta_plus - beta_minus
            # Objective: min 1^T beta_plus + 1^T beta_minus
            # Subject to: X beta = y
            fit = linprog(
                np.ones(2 * p),
                A_eq=np.concatenate([X, -X], axis=1),
                b_eq=y,
                bounds=(0, None),
                method="highs",
            )

            if fit.success:
                beta_hat = fit.x[:p] - fit.x[p:]

                if np.linalg.norm(beta_hat - beta) <= 1e-4 * np.linalg.
 ↪norm(beta):
                    successes += 1

    return i, j, successes / T

rng = np.random.SeedSequence(0)
seeds = rng.spawn(len(n_values) * len(k_values))

n_jobs = 8

# Parallelize across (n, k) configurations
results = Parallel(
    n_jobs=n_jobs,
    backend="loky",
    verbose=10
```

```
)(
    delayed(run_config)(i, j, n, k, seeds[i * len(k_values) + j])
    for i, n in enumerate(n_values)
    for j, k in enumerate(k_values)
)

# Reconstruct success probability matrix
success = np.zeros((len(n_values), len(k_values)))

for i, j, rate in results:
    success[i, j] = rate
```

```
[Parallel(n_jobs=8)]: Using backend LokyBackend with 8 concurrent workers.
[Parallel(n_jobs=8)]: Done   2 tasks      | elapsed:    1.2s
[Parallel(n_jobs=8)]: Done   9 tasks      | elapsed:    1.3s
[Parallel(n_jobs=8)]: Done  16 tasks      | elapsed:    1.3s
[Parallel(n_jobs=8)]: Done  25 tasks      | elapsed:    1.5s
[Parallel(n_jobs=8)]: Done  34 tasks      | elapsed:    1.6s
[Parallel(n_jobs=8)]: Done  45 tasks      | elapsed:    1.8s
[Parallel(n_jobs=8)]: Done  56 tasks      | elapsed:    1.9s
[Parallel(n_jobs=8)]: Done  69 tasks      | elapsed:    2.2s
[Parallel(n_jobs=8)]: Done  82 tasks      | elapsed:    2.6s
[Parallel(n_jobs=8)]: Done  97 tasks      | elapsed:    3.0s
[Parallel(n_jobs=8)]: Done 112 tasks      | elapsed:    3.3s
[Parallel(n_jobs=8)]: Done 129 tasks      | elapsed:    3.8s
[Parallel(n_jobs=8)]: Done 146 tasks      | elapsed:    4.4s
[Parallel(n_jobs=8)]: Done 165 tasks      | elapsed:    5.1s
[Parallel(n_jobs=8)]: Done 184 tasks      | elapsed:    5.9s
[Parallel(n_jobs=8)]: Done 205 tasks      | elapsed:    6.8s
[Parallel(n_jobs=8)]: Done 226 tasks      | elapsed:    7.9s
[Parallel(n_jobs=8)]: Done 249 tasks      | elapsed:    9.1s
[Parallel(n_jobs=8)]: Done 272 tasks      | elapsed:   10.6s
[Parallel(n_jobs=8)]: Done 297 tasks      | elapsed:   12.3s
[Parallel(n_jobs=8)]: Done 322 tasks      | elapsed:   14.1s
[Parallel(n_jobs=8)]: Done 349 tasks      | elapsed:   18.3s
[Parallel(n_jobs=8)]: Done 376 tasks      | elapsed:   20.7s
[Parallel(n_jobs=8)]: Done 405 tasks      | elapsed:   23.6s
[Parallel(n_jobs=8)]: Done 434 tasks      | elapsed:   31.8s
[Parallel(n_jobs=8)]: Done 465 tasks      | elapsed:   45.3s
[Parallel(n_jobs=8)]: Done 496 tasks      | elapsed:  1.0min
[Parallel(n_jobs=8)]: Done 529 tasks      | elapsed:  1.1min
[Parallel(n_jobs=8)]: Done 562 tasks      | elapsed:  1.2min
[Parallel(n_jobs=8)]: Done 597 tasks      | elapsed:  1.3min
[Parallel(n_jobs=8)]: Done 632 tasks      | elapsed:  1.5min
[Parallel(n_jobs=8)]: Done 669 tasks      | elapsed:  1.8min
[Parallel(n_jobs=8)]: Done 706 tasks      | elapsed:  1.9min
[Parallel(n_jobs=8)]: Done 745 tasks      | elapsed:  2.0min
[Parallel(n_jobs=8)]: Done 784 tasks      | elapsed:  2.1min
```

```
[Parallel(n_jobs=8)]: Done 825 tasks      | elapsed:  2.3min
[Parallel(n_jobs=8)]: Done 866 tasks      | elapsed:  2.4min
[Parallel(n_jobs=8)]: Done 909 tasks      | elapsed:  2.6min
[Parallel(n_jobs=8)]: Done 952 tasks      | elapsed:  2.8min
[Parallel(n_jobs=8)]: Done 997 tasks      | elapsed:  3.0min
[Parallel(n_jobs=8)]: Done 1024 out of 1024 | elapsed:  3.1min finished
```

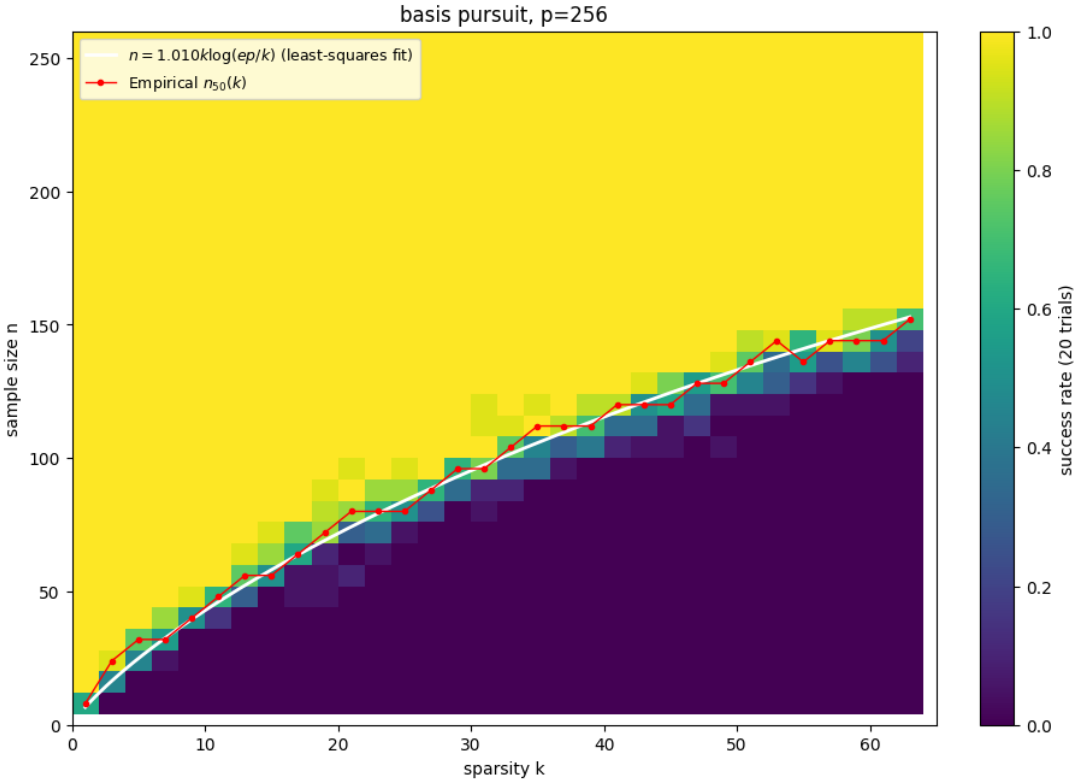
[17]:
```
np.save("success.npy", success)
```

[19]:
```
n50 = np.array([n_values[np.where(success[:,j] >= 0.5)[0][0]] if np.
 ↪any(success[:,j]>=0.5)
                else np.nan for j in range(len(k_values))])
factor = k_values * np.log(np.e * p / k_values)
valid = np.isfinite(n50)
C = np.sum(factor[valid]*n50[valid]) / np.sum(factor[valid]**2)

fig, ax = plt.subplots(figsize=(9,6.5), layout='constrained')
k_edges = np.r_[k_values[0] - (k_values[1] - k_values[0]) / 2, (k_values[:-1] +␣
 ↪k_values[1:]) / 2,
                k_values[-1] + (k_values[-1] - k_values[-2]) / 2]
n_edges = np.r_[n_values[0] - (n_values[1] - n_values[0]) / 2, (n_values[:-1] +␣
 ↪n_values[1:]) / 2,
                n_values[-1] + (n_values[-1] - n_values[-2]) / 2]
heat = ax.pcolormesh(k_edges, n_edges, success, vmin=0, vmax=1, cmap='viridis',␣
 ↪shading='flat')
k_dense = np.linspace(1, 63, 400)
ax.plot(k_dense, C * k_dense * np.log(np.e * p / k_dense), color='white', lw=2,
        label=rf'$n = {C:.3f} k\log(ep/k)$ (least-squares fit)')
ax.plot(k_values[valid], n50[valid], color='red', marker='o', lw=1,␣
 ↪markersize=3,
        label=r'Empirical $n_{50}(k)$')
ax.set(xlabel='sparsity k', ylabel='sample size n', title='basis pursuit,␣
 ↪p=256')
ax.set_xlim(0, 65)
ax.set_ylim(0, 260)
ax.legend(loc='upper left', fontsize=9)
cbar = fig.colorbar(heat, ax=ax)
cbar.set_label(f'success rate ({T} trials)')
print('C:',C)
print('n50:',list(zip(k_values,n50)))
```

```
C: 1.010011803526003
n50: [(np.int64(1), np.int64(8)), (np.int64(3), np.int64(24)), (np.int64(5),
np.int64(32)), (np.int64(7), np.int64(32)), (np.int64(9), np.int64(40)),
(np.int64(11), np.int64(48)), (np.int64(13), np.int64(56)), (np.int64(15),
np.int64(56)), (np.int64(17), np.int64(64)), (np.int64(19), np.int64(72)),
(np.int64(21), np.int64(80)), (np.int64(23), np.int64(80)), (np.int64(25),
np.int64(80)), (np.int64(27), np.int64(88)), (np.int64(29), np.int64(96)),
```

```
(np.int64(31), np.int64(96)), (np.int64(33), np.int64(104)), (np.int64(35),
np.int64(112)), (np.int64(37), np.int64(112)), (np.int64(39), np.int64(112)),
(np.int64(41), np.int64(120)), (np.int64(43), np.int64(120)), (np.int64(45),
np.int64(120)), (np.int64(47), np.int64(128)), (np.int64(49), np.int64(128)),
(np.int64(51), np.int64(136)), (np.int64(53), np.int64(144)), (np.int64(55),
np.int64(136)), (np.int64(57), np.int64(144)), (np.int64(59), np.int64(144)),
(np.int64(61), np.int64(144)), (np.int64(63), np.int64(152))]
```



i. The shape of the empirical boundary (red dotted lines) follows $k\log(ep/k)$, $C = 1.01$.

ii. The result predicts the shape of the boundary because the required sample size scales proportionally to $k\log(ep/k)$. However, the theorem doesn't specify the exact constant C', so it cannot determine the precise sample size at which exact recovery becomes likely. Furthermore, the condition provides a sufficient condition for exact recovery, not the threshold for 50% recovery as in i., thus the location is never precisely known. The transition is sharp because Gaussian random matrices exhibit strong concentration properties. Below the critical sample size, basis pursuit often can't uniquely recover the sparse signal, but once n becomes sufficiently large relative to $k\log(ep/k)$, the necessary conditions for exact recovery take root. Consequently, the success probability is sharp.

1.3 (c)

[7]:
```
from sklearn.linear_model import Lasso

p = 512
sigma = 1
k = 10
n_values = np.array([200, 400, 800, 1600, 3200])
k_values = np.array([5, 10, 15, 20, 25, 30])
configs = [(int(n), 10) for n in n_values] + [(800, int(k)) for k in k_values␣
 ↪if k != 10]

def run_config(n, k, seed):
    rng = np.random.default_rng(seed)
    values = []
    with threadpool_limits(limits=1):
        for _ in range(T):
            X = rng.standard_normal((n, p))
            beta = np.zeros(p)
            support = rng.choice(p, size=k, replace=False)
            beta[support] = rng.choice([-1.0, 1.0], size=k)
            eps = rng.normal(0, sigma, size=n)
            y = X @ beta + eps
            lam = 0.5 * sigma * np.sqrt(2 * np.log(p) / n)
            fit = Lasso(alpha=lam, fit_intercept=False, max_iter=20000,␣
 ↪tol=1e-7)
            fit.fit(X, y)
            values.append(np.sum((fit.coef_ - beta) ** 2))
    values = np.asarray(values)
    return (n, k, float(values.mean()), float(values.std(ddof=1) / np.sqrt(T)),␣
 ↪float(values.std(ddof=1)))

n_jobs = 8
seeds = np.random.SeedSequence(0).spawn(len(configs))
results = Parallel(n_jobs=n_jobs, backend='loky', verbose=10)(
    delayed(run_config)(n, k, seeds[i]) for i, (n, k) in enumerate(configs)
)

for n, k, mean, se, sd in results:
    print(f"n={n}, k={k}, mean MSE={mean:.4f}, SE={se:.4f}, SD={sd:.4f}",␣
 ↪flush=True)
np.savez("results_3c.npz", results=np.array(results), p=p, sigma=sigma, T=T,␣
 ↪n_values=n_values, k_values=k_values)
```

```
[Parallel(n_jobs=8)]: Using backend LokyBackend with 8 concurrent workers.
[Parallel(n_jobs=8)]: Done   3 out of  10 | elapsed:    2.7s remaining:    6.3s
[Parallel(n_jobs=8)]: Done   5 out of  10 | elapsed:    2.7s remaining:    2.7s
[Parallel(n_jobs=8)]: Done   7 out of  10 | elapsed:    2.8s remaining:    1.1s
```

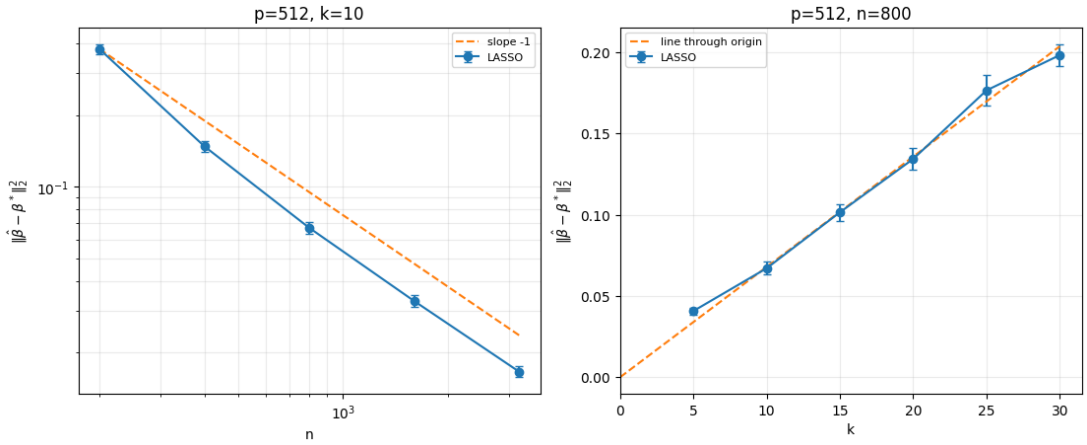
```
n=200, k=10, mean MSE=0.3790, SE=0.0192, SD=0.0857
n=400, k=10, mean MSE=0.1477, SE=0.0078, SD=0.0351
n=800, k=10, mean MSE=0.0670, SE=0.0040, SD=0.0180
n=1600, k=10, mean MSE=0.0330, SE=0.0019, SD=0.0085
n=3200, k=10, mean MSE=0.0166, SE=0.0009, SD=0.0039
n=800, k=5, mean MSE=0.0407, SE=0.0022, SD=0.0098
n=800, k=15, mean MSE=0.1015, SE=0.0052, SD=0.0230
n=800, k=20, mean MSE=0.1343, SE=0.0068, SD=0.0303
n=800, k=25, mean MSE=0.1765, SE=0.0092, SD=0.0412
n=800, k=30, mean MSE=0.1982, SE=0.0069, SD=0.0308

[Parallel(n_jobs=8)]: Done  10 out of  10 | elapsed:    3.3s finished
```

[10]:

```
lookup = {(int(n), int(k)):(m,se) for n, k, m, se, _ in results}
mean_n = np.array([lookup[(int(n), 10)][0] for n in n_values])
sem_n = np.array([lookup[(int(n), 10)][1] for n in n_values])
mean_k = np.array([lookup[(800, int(k))][0] for k in k_values])
sem_k = np.array([lookup[(800, int(k))][1] for k in k_values])
slope, intercept = np.polyfit(np.log(n_values), np.log(mean_n), 1)
linear_coefficient = np.sum(k_values * mean_k) / np.sum(k_values**2)
print(f'loglog slope = {slope:.6f}; slope through origin (k) =␣
 ↪{linear_coefficient:.6f}')
fig, axs = plt.subplots(1, 2, figsize=(11, 4.5), layout='constrained')
axs[0].errorbar(n_values, mean_n, yerr=sem_n, fmt='o-', capsize=3,␣
 ↪label='LASSO')
axs[0].plot(n_values, mean_n[0] * n_values[0] / n_values, '--', label='slope␣
 ↪-1')
axs[0].set(xscale='log', yscale='log', xlabel='n', ylabel=r"$\|\hat{\beta} -␣
 ↪\beta^*\|^2_2$", title='p=512, k=10')
axs[0].legend(fontsize=8)
axs[0].grid(True, alpha=.25, which='both')
axs[1].errorbar(k_values, mean_k, yerr=sem_k, fmt='o-', capsize=3,␣
 ↪label='LASSO')
axs[1].plot([0, *k_values], [0, *(linear_coefficient * k_values)], '--',␣
 ↪label='line through origin')
axs[1].set(xlabel='k', ylabel=r"$\|\hat{\beta} - \beta^*\|^2_2$", title='p=512,␣
 ↪n=800')
axs[1].set_xlim(left=0)
axs[1].legend(fontsize=8)
axs[1].grid(True, alpha=.25, which='both')
plt.show()
```

```
loglog slope = -1.118073; slope through origin (k) = 0.006786
```



i. The loglog slope is -1.12, which is close to the theoretical slope of -1. From lecture 8, the squared estimation error is bounded by $\frac{\sigma^2 k \log p}{n}$, so for a fixed $k = 10$, $p = 512$, and $\sigma = 1$, this predicts an inverse relationship with n, which we see in the graph on the left. The differences arise from Monte Carlo variability / random seed set.

ii. Yes, there is a linear dependence on k in $\frac{\sigma^2 k \log p}{n}$, as $p = 512$, $n = 800$, and $\sigma = 1$ (all fixed values). This is exemplified in the graph on the right, as the error increases roughly linearly as k increases.

1.4 (d)

[12]:
```
p_k_values = [(128, 12), (256, 16), (512, 23), (1024, 32)]
r_values = np.array([1, 2, 3, 4, 6, 8, 12, 16])
configs = [(p, k, float(r), int(round(r * k * np.log(p)))) for p, k in␣
↪p_k_values for r in r_values]

def run_config(p, k, r, n, seed):
    rng = np.random.default_rng(seed)
    sqerr = np.empty(T)
    lam = 0.5 * np.sqrt(2 * np.log(p) / n)
    with threadpool_limits(limits=1):
        for t in range(T):
            X = rng.standard_normal((n, p))
            beta = np.zeros(p)
            support = rng.choice(p, size=k, replace=False)
            beta[support] = rng.choice([-1.0, 1.0], size=k)
            eps = rng.normal(0, 1, size=n)
            y = X @ beta + eps
            fit = Lasso(alpha=lam, fit_intercept=False, max_iter=20000,␣
↪tol=1e-7)
            fit.fit(X, y)
```

```
            sqerr[t] = np.sum((fit.coef_ - beta) ** 2)
    return p, k, r, n, float(sqerr.mean()), float(sqerr.std(ddof=1) / np.
 ↪sqrt(T)), float(sqerr.std(ddof=1))

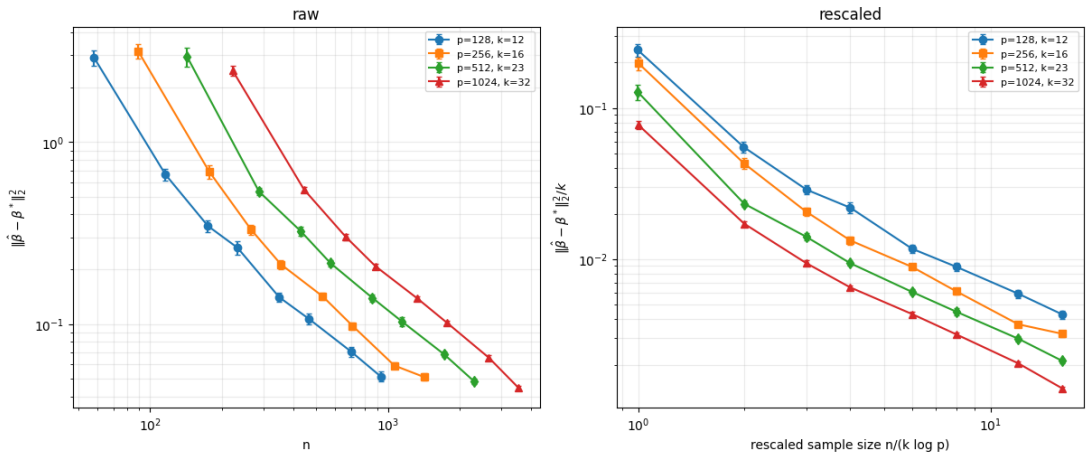
n_jobs = 8
seeds = np.random.SeedSequence(0).spawn(len(configs))
results = Parallel(n_jobs=n_jobs, backend='loky', verbose=10)(
    delayed(run_config)(p, k, r, n, seeds[i])
    for i, (p, k, r, n) in enumerate(configs)
)
np.savez("results_3d.npz", results=np.array(results), p_k_values=np.
 ↪asarray(p_k_values), r_values=r_values, T=T, sigma=sigma)
```

```
[Parallel(n_jobs=8)]: Using backend LokyBackend with 8 concurrent workers.
[Parallel(n_jobs=8)]: Done   2 tasks      | elapsed:    2.5s
[Parallel(n_jobs=8)]: Done   9 tasks      | elapsed:    2.6s
[Parallel(n_jobs=8)]: Done  16 tasks      | elapsed:    2.7s
[Parallel(n_jobs=8)]: Done  21 out of  32 | elapsed:    3.0s remaining:    1.5s
[Parallel(n_jobs=8)]: Done  25 out of  32 | elapsed:    3.3s remaining:    0.8s
[Parallel(n_jobs=8)]: Done  29 out of  32 | elapsed:    3.9s remaining:    0.3s
[Parallel(n_jobs=8)]: Done  32 out of  32 | elapsed:    5.4s finished
```

[18]:
```
results = np.array(results)
for p, k, r, n, mean, sem, _ in results:
    print(f'p={p:4.0f} k={k:2.0f} r={r:2.0f} n={n:4.0f} mean={mean:.6f}␣
 ↪SEM={sem:.6f}')
fig, axs = plt.subplots(1, 2, figsize=(12, 5), layout='constrained')
fmt_values = ['o-', 's-', 'd-', '^-']
for i, (p, k) in enumerate(p_k_values):
    rows = results[(results[:, 0]==p) & (results[:, 1]==k)]
    n, means, sem = rows[:, 3], rows[:, 4], rows[:, 5]
    r = rows[:, 3] / (k * np.log(p))
    axs[0].errorbar(n, means, yerr=sem, fmt=fmt_values[i], capsize=2,␣
 ↪label=f'p={p}, k={k}')
    axs[1].errorbar(r ,means / k, yerr=sem / k, fmt=fmt_values[i], capsize=2,␣
 ↪label=f'p={p}, k={k}')
axs[0].set(xscale='log', yscale='log', xlabel='n', ylabel=r"$\|\hat{\beta} -␣
 ↪\beta^*\|^2_2$", title='raw')
axs[1].set(xscale='log', yscale='log', xlabel='rescaled sample size n/(k log␣
 ↪p)', ylabel=r"$\|\hat{\beta} - \beta^*\|^2_2 / k$", title='rescaled')
axs[0].legend(fontsize=8)
axs[0].grid(True, alpha=.25, which='both')
axs[1].legend(fontsize=8)
axs[1].grid(True, alpha=.25, which='both')
plt.show()
```

```
p= 128 k=12 r= 1 n=  58 mean=2.914070 SEM=0.276783
p= 128 k=12 r= 2 n= 116 mean=0.662994 SEM=0.050634
```

```
p= 128 k=12 r= 3 n= 175 mean=0.346022 SEM=0.024619
p= 128 k=12 r= 4 n= 233 mean=0.262795 SEM=0.021294
p= 128 k=12 r= 6 n= 349 mean=0.140581 SEM=0.008068
p= 128 k=12 r= 8 n= 466 mean=0.106751 SEM=0.007058
p= 128 k=12 r=12 n= 699 mean=0.070793 SEM=0.004387
p= 128 k=12 r=16 n= 932 mean=0.051654 SEM=0.003269
p= 256 k=16 r= 1 n=  89 mean=3.158824 SEM=0.299729
p= 256 k=16 r= 2 n= 177 mean=0.689777 SEM=0.058092
p= 256 k=16 r= 3 n= 266 mean=0.330009 SEM=0.019710
p= 256 k=16 r= 4 n= 355 mean=0.212843 SEM=0.012672
p= 256 k=16 r= 6 n= 532 mean=0.142003 SEM=0.006725
p= 256 k=16 r= 8 n= 710 mean=0.097944 SEM=0.004612
p= 256 k=16 r=12 n=1065 mean=0.059240 SEM=0.002601
p= 256 k=16 r=16 n=1420 mean=0.051249 SEM=0.002415
p= 512 k=23 r= 1 n= 143 mean=2.944675 SEM=0.349473
p= 512 k=23 r= 2 n= 287 mean=0.534898 SEM=0.024375
p= 512 k=23 r= 3 n= 430 mean=0.323961 SEM=0.019165
p= 512 k=23 r= 4 n= 574 mean=0.216099 SEM=0.007800
p= 512 k=23 r= 6 n= 861 mean=0.139295 SEM=0.005905
p= 512 k=23 r= 8 n=1148 mean=0.103211 SEM=0.006074
p= 512 k=23 r=12 n=1722 mean=0.068360 SEM=0.002669
p= 512 k=23 r=16 n=2296 mean=0.048740 SEM=0.001742
p=1024 k=32 r= 1 n= 222 mean=2.476591 SEM=0.152731
p=1024 k=32 r= 2 n= 444 mean=0.548146 SEM=0.021855
p=1024 k=32 r= 3 n= 665 mean=0.301501 SEM=0.012202
p=1024 k=32 r= 4 n= 887 mean=0.207954 SEM=0.006613
p=1024 k=32 r= 6 n=1331 mean=0.138339 SEM=0.004403
p=1024 k=32 r= 8 n=1774 mean=0.101693 SEM=0.003244
p=1024 k=32 r=12 n=2662 mean=0.065323 SEM=0.002066
p=1024 k=32 r=16 n=3549 mean=0.044595 SEM=0.001308
```



The raw error curves do not collapse when plotted against n because the estimation error depends

on the sparsity k and the ambient dimension p, in addition to the sample size. Rescaling the horizontal axis by $k \log p$ accounts for the sample complexity in the bound, causing the curves to align more closely in shape. However, the vertical rescaling do not perfectly collapse because the squared error bound predicts with a remaining factor of $1/k$, which is why we still see some separation. This shows that the ambient dimension p enters the problem of sample complexity only logarithmically, while the sparsity k enteres linearly. Consequently, increasing the number of features has a much weaker effect on the required sample size than increasing the number of coefficients.

1.5 (e)

[25]:
```
p = 512
k = 10
n_values = np.array([200, 400, 800])
c_values = np.geomspace(0.03, 3, 11)

def run_config(n, seed):
    rng = np.random.default_rng(seed)
    X = rng.standard_normal((n, p))
    beta = np.zeros(p)
    support = rng.choice(p, size=k, replace=False)
    beta[support] = rng.choice([-1.0, 1.0], size=k)
    y = X @ beta + rng.normal(0, sigma, size=n)
    lambda0 = sigma * np.sqrt(2 * np.log(p) / n)
    errors = np.empty(len(c_values))
    with threadpool_limits(limits=1):
        for i, c in enumerate(c_values):
            fit = Lasso(alpha=float(c * lambda0), fit_intercept=False,␣
 ↪max_iter=20000, tol=1e-7)
            fit.fit(X, y)
            errors[i] = np.sum((fit.coef_ - beta) ** 2)
    return errors

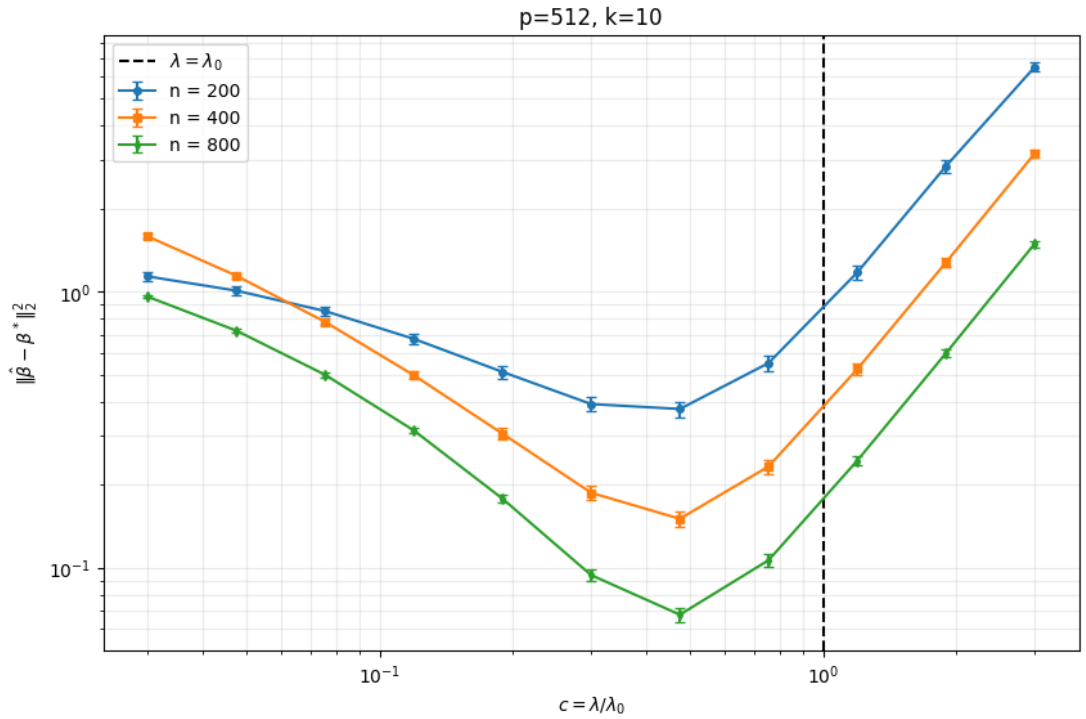
configs = [(i, int(n), t) for i, n in enumerate(n_values) for t in range(T)]
seeds = np.random.SeedSequence(0).spawn(len(configs))
results = Parallel(n_jobs=8, backend='loky', verbose=10)(
    delayed(run_config)(n, seeds[i * T + t]) for i, n, t in configs
)
errors = np.asarray(results).reshape(len(n_values), T, len(c_values))
means = errors.mean(axis=1)
sems = errors.std(axis=1, ddof=1) / np.sqrt(T)
np.savez("results_3e.npz", errors=errors, means=means, sems=sems,␣
 ↪n_values=n_values, c_values=c_values, p=p, k=k, T=T, sigma=sigma)
```

```
[Parallel(n_jobs=8)]: Using backend LokyBackend with 8 concurrent workers.
[Parallel(n_jobs=8)]: Done   2 tasks      | elapsed:    2.6s
[Parallel(n_jobs=8)]: Done   9 tasks      | elapsed:    2.8s
[Parallel(n_jobs=8)]: Done  16 tasks      | elapsed:    2.9s
```

```
[Parallel(n_jobs=8)]: Done  25 tasks      | elapsed:    3.0s
[Parallel(n_jobs=8)]: Done  34 tasks      | elapsed:    3.2s
[Parallel(n_jobs=8)]: Done  45 tasks      | elapsed:    3.4s
[Parallel(n_jobs=8)]: Done  52 out of  60 | elapsed:    3.5s remaining:    0.5s
[Parallel(n_jobs=8)]: Done  60 out of  60 | elapsed:    3.6s finished
```

[28]:
```
fig, ax = plt.subplots(figsize=(8.4, 5.6), layout='constrained')
fmt_values = ['o-', 's-', 'd-']
for i, n in enumerate(n_values):
    ax.errorbar(c_values, means[i], yerr=sems[i], fmt=fmt_values[i], capsize=3,
                linewidth=1.5, markersize=4, label=f'n = {n}')
ax.axvline(1.0, linestyle='--', color='black', label=r'$\lambda=\lambda_0$')
ax.set(xscale='log', yscale='log', xlabel=r'$c=\lambda/\lambda_0$',
       ylabel=r"$\|\hat{\beta} - \beta^*\|^2_2$",
       title='p=512, k=10')
ax.grid(True, which='both', alpha=0.25)
ax.legend()
print('Grid c:', c_values)
for i, n in enumerate(n_values):
    j = np.argmin(means[i]);print(f'n={n}, optimum c={c_values[j]:.5f}, mean␣
 ↪error={means[i,j]:.6f}')
    print('means:',np.round(means[i],6))
plt.show()
```

```
Grid c: [0.03       0.0475468  0.07535659 0.11943215 0.1892872  0.3
 0.47546796 0.75356593 1.19432151 1.89287203 3.        ]
n=200, optimum c=0.47547, mean error=0.376258
means: [1.136864 1.009343 0.851425 0.675207 0.512646 0.392038 0.376258 0.551893
 1.174782 2.849869 6.513563]
n=400, optimum c=0.47547, mean error=0.150624
means: [1.586634 1.143507 0.778247 0.498768 0.306356 0.187052 0.150624 0.232959
 0.524424 1.277086 3.166984]
n=800, optimum c=0.47547, mean error=0.067693
means: [0.959803 0.72317  0.500575 0.314795 0.178131 0.094166 0.067693 0.106594
 0.244575 0.599161 1.492343]
```



The minimum occurs around $0.475\lambda_0$, which is close to the $\lambda_0/2$ used in parts (c) and (d). When λ is too small, the ℓ_1 penalty is weak, allowing the LASSO to fit noise and include unecessary variables, which increases the variance and estimation error. When λ is too large, the penalty becomes too strong, excessively shrinking the nonzero coefficients toward zero, which increases bias and estimation error. This is why we see a U-shaped curve in the graph, visualizing the bias-variance tradeoff, and also why the lowest errors occur in middle λ values. Our empirically optimal λ in (a) was $0.92\lambda_0$. From the proof in lecture 7, this would result in $1.84\lambda_0$, which is much larger than the empirically optimal regularization level we got of $0.475\lambda_0$. This difference is because the theoretical condition is designed to guarantee a high-probability error bound, ensuring the noise term is controlled uniformly, whereas the simulation minimizes average estimation error so a smaller value of λ can perform better empirically.

AI Usage: All homework problems were written in VSCode and compiled using LaTeX (Problem 3 utilized Jupyter notebook), with assistance from ChatGPT and Github Copilot inline suggestions.